A Linear Analysis of the Heating Footprints of the U.S. Mesoscale Convective Systems in the Northern Extratropical Synoptic Variability

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(Manuscript received 22 September 2024, in final form 1 April 2025, accepted 28 April 2025)

ABSTRACT: This study employs a two-layer quasigeostrophic (QG) channel model with a linear, empirical parameterization of diabatic heating to investigate the impact of the U.S. mesoscale convective systems (MCSs) on the northern extratropical synoptic variability in boreal spring. Major modes of synoptic variability in the model are obtained from fast-growing "dry" and "moist" optimal disturbances developing in a climatological spring background flow. Compared to the no-heating scenario, the introduction of domainwide diabatic heating generally reduces the zonal scale and increases the amplitude of optimal disturbances, leading to two better-defined maxima of synoptic-scale variance corresponding to the North Pacific and North Atlantic storm tracks. Dominated by MCS heating, diabatic heating over the central United States not only significantly enhances synoptic variability over the eastern United States and North Atlantic but also promotes the circumglobal zonal propagation of synoptic-scale disturbances, extending its impact to planetary and hemispheric scales. The transient (synoptic) eddy forcing of the background zonal flow is notably amplified by MCS heating, manifested as more prominent westerly accelerations that strengthen and extend the Atlantic jet downstream. The implications of the findings from this highly idealized analysis are further discussed in the context of the upscale feedback of mesoscale disturbances to extratropical low-frequency variability and the coupling between long-term trends in U.S. MCS activity and regional and hemispheric circulation changes.

SIGNIFICANCE STATEMENT: This study investigates how localized heating from mesoscale storms (mesoscale convective systems over the United States) affects weather patterns across vast regions. Using an idealized model, this research shows that convective heating from these storms can extend its influence far beyond the United States, potentially altering weather patterns over the North Atlantic and Eurasian continents for days or even longer. In addition, latent heat release from these storms can energize weather systems operating on much longer time scales and thus have broad implications for regional and global climate. This research underscores the need for further studies of the interactions between local storm activity and global weather and climate patterns, ultimately contributing to improved predictions and response strategies to adverse climate conditions.

KEYWORDS: Heating; Instability; Large-scale motions; Mesoscale systems; Quasigeostrophic models; Climate variability

1. Introduction

Mesoscale convective systems (MCSs), among the largest convective storms, develop when aggregated convections organize into distinct mesoscale circulations (Houze 2004). In the United States, particularly east of the Rocky Mountains, MCSs are notably active during the boreal spring (March–May), producing severe weather hazards including strong straight-line winds, hail, and tornadoes (Bentley and Sparks 2003; Houze et al. 1990; Smith et al. 2012). These systems are significant contributors to precipitation, accounting for 50%–70% of the total spring rainfall and 60%–75% of the region's extreme rainfall events over the central Great Plains (Schumacher and Johnson 2006). MCSs are regulated by forcing mechanisms that operate across multiple spatial and temporal scales. At the regional scale, the exact locations and life cycles of MCSs are determined by regional circulations and thermodynamic factors such as low-level jets, fronts, and convective instability (Coniglio et al. 2010; Song et al. 2019; Feng et al. 2019; You et al. 2024). These local factors provide favorable conditions that are critical in the development and maturation of MCSs. Moving to larger scales, MCSs are typically triggered by ascending motions ahead of large-scale troughs, whose formations can be attributed to local instability or remote forcing of various origins (You and Deng 2023). These large-scale forcing mechanisms control the synoptic to subseasonal MCS variability as well as the long-term trend in MCS genesis frequency across the United States (Feng et al. 2016; You and Deng 2023).

Despite extensive research on MCSs and their associated multiscale forcing mechanisms, studies examining the feedback of MCSs on their large-scale forcing remain limited. MCSs are known to influence synoptic-scale circulations through their top-heavy heating profile. The generation of potential vorticity (PV), which is proportional to the vertical gradient of heating, enables MCS-induced heating to inject PV into the midtroposphere, creating what is commonly referred to as a "mesoscale convective vortex" (MCV) (Bartels and Maddox 1991). This MCV can amplify local synoptic-scale troughs, thereby extending the MCS's life cycle and enhancing its precipitation efficiency (Trier and Davis 2002; Schumacher et al. 2013). However, the broader

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DOI: 10.1175/JCLI-D-24-0543.1

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implications of MCS heating for large-scale circulations, particularly across planetary and hemispheric scales, remain largely unexplored. From a dynamical perspective, comprehending how regional convective systems influence synoptic- and planetary-scale circulations is crucial for addressing knowledge gaps in our understanding of the scale interactions within the atmosphere. Such knowledge can also help identify sources of predictability for synoptic to subseasonal circulation anomalies and associated weather extremes over regions far away from MCS genesis locations. This is particularly important given that warm-season MCS events have become more frequent over the recent decades (Feng et al. 2016), and this increasing trend in MCS activity is expected to continue in a warming climate (Prein et al. 2017; Song et al. 2022; Cui et al. 2024).

To extract and delineate the feedback of MCSs on large-scale circulations, it is essential to employ a dynamical system that integrates the effects of MCS heating, especially latent heating. The complex thermodynamics of moist air (Emanuel 1994) and the nonlinear relationship between latent heat release and large-scale atmospheric circulations make it much more challenging to study moist baroclinic instability compared to its dry counterpart. In general, moisture is known to reduce the static stability of atmospheric flow (Emanuel et al. 1987; Lapeyre and Held 2004; O'Gorman 2011; Booth et al. 2015). Early theoretical studies on moist baroclinic instability have employed parameterizations associated with the conditional instability of the second kind (CISK) (Mak 1982; Shukla 1978), which links latent heat release to low-level convergence within the surface moist layer (Charney and Eliassen 1964). Other studies have adopted a different approach, assuming diabatic heating to be proportional to the large-scale vertical motion at every level, which can be further modulated into varying stratification between ascending and descending regions (Emanuel et al. 1987; Whitaker and Davis 1994; Montgomery and Farrell 1991). Numerical simulations, using both quasigeostrophic (QG) (Emanuel et al. 1987) and primitive equations (Whitaker and Davis 1994), indicate an increased growth rate of synoptic disturbances due to condensational heating. Despite these findings, the effort to explore moist eddy growth from a nonmodal perspective remains limited, often constrained by the nonlinear nature of the parameterizations of latent heat release. Moreover, the impact of *regional* heating, particularly in the MCS-active region over the central United States, on large-scale disturbances across the Northern Hemispheric midlatitudes has not been extensively studied in the context of an idealized model. Results obtained from such analyses hold key insights for assessing and understanding the impacts of changing moisture content and mesoscale convective processes on the life cycles of synoptic- to hemispheric-scale circulations and their implications for global climate change.

Employing a two-layer QG model with a linear empirical parameterization of diabatic heating, this study analyzes the feedback of regional (United States) MCS heating on northern extratropical synoptic variability in a channel domain. Section 2 introduces the dynamical equations used in the two-layer QG model, the parameterization of diabatic heating, and the method of nonmodal instability analysis. Section 3 presents first the general results of the moist nonmodal instability analysis, followed by a discussion of the downstream effects of MCS heating on synoptic-scale disturbances as well as their feedback to the seasonal mean flow over the North America–North Atlantic sector. Section 4 provides some concluding remarks for this idealized analysis.

2. Methodology

a. Moist QG potential vorticity equations

We derive the moist QGPV equations (i.e., with diabatic heating) for the two-layer model starting from the quasigeostrophic vorticity and thermodynamic equations (Mak 2011).

First, applying geostrophic balance and the continuity equation to the quasigeostrophic vorticity equation and employing vertical boundary conditions, we express the vorticity equations for the upper and lower levels [i.e., 250 and 750 mb (1 mb = 100 hPa)] as follows:

$$\frac{\partial \zeta_1}{\partial t} + f^{-1}J(Z_1, \zeta_1 + f) = f\frac{\omega_2}{\delta p} - \epsilon_1\zeta_1, \quad \frac{\partial \zeta_3}{\partial t} + f^{-1}J(Z_3, \zeta_3 + f) = -f\frac{\omega_2}{\delta p} - \epsilon_3\zeta_3. \tag{1}$$

In these equations, subscripts 1, 2, and 3 denote the upper, mid- (500 mb), and lower levels, respectively. The Jacobian operator $J(A, B) = A_xB_y - A_yB_x$ represents the horizontal advection terms. Vorticity ζ and geopotential Z are related to each other by the equation $\zeta = f^{-1}\nabla^2 Z$ in the QG system. $f = 2\Omega\sin\varphi$ is the Coriolis parameter with Ω being the angular velocity of Earth's self-rotation and φ is the latitude. $\delta p = 500$ mb represents the thickness of the layer and ω denotes the vertical velocity. Linear damping on relative vorticity is employed in the model to mimic the synoptic time scale spindown effect associated with the presence of a planetary boundary layer. In this study, the damping magnitude is set at $\varepsilon_3 = (5\ \text{day})^{-1}$ at the lower level and is reduced to $\varepsilon_1 = (10\ \text{day})^{-1}$ at the upper level to reflect the diminishing influence of boundary layer processes at higher altitudes.

Next, the QG thermodynamic equation at the midlevel, which includes the effect of diabatic heating, is given by

$$\frac{\partial}{\partial t}\left[\frac{(Z_1 - Z_3)}{\delta p}\right] + \frac{1}{f\delta p}J(Z_3, Z_1) = \omega_2 S_2 + \frac{R}{C_p p_2}Q. \tag{2}$$

Here, S_2 is the horizontally varying static stability parameter ($\text{N}^{-2}\ \text{m}^6\ \text{s}^{-2}$) in the QG system, Q is the local diabatic heating rate ($\text{J kg}^{-1}\ \text{s}^{-1}$), $p_2 = 500$ mb is the pressure at the midlevel, $C_p = 1004\ \text{J kg}^{-1}\ \text{K}^{-1}$ is the specific heat capacity at constant pressure, and $R = 287\ \text{J kg}^{-1}\ \text{K}^{-1}$ is the gas constant for dry air. The diabatic heating term appears as the second term on the right-hand side of the equation. The scaling coefficient R/p_2C_p of the heating term arises from the constraint of hydrostatic balance within the QG framework when deriving the thermodynamic equation in a pressure coordinate.

To incorporate diabatic heating into the QG system, we first estimate the local diabatic heating rate at the midlevel using the residual method (e.g., Nigam 1994):

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$$Q = \frac{C_p T}{\theta}\left(\frac{\partial \theta}{\partial t} + u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} + \omega\frac{\partial \theta}{\partial p}\right), \tag{3}$$

where θ is the local potential temperature, T is the local temperature, and u and v are zonal and meridional winds, respectively.

Subsequently, we train a linear regression model where local Q is regressed onto local ω_2 *at each grid point* without intercept:

$$Q = \alpha\omega_2. \tag{4}$$

In the major heating regions of the model domain (e.g., storm-track regions, central United States), the diabatic heating at the midlevel (500 mb) is predominantly contributed by latent heating (not shown). A critical assumption in our analysis is that strong latent heat release (positive Q) is typically associated with *local* ascent (negative ω). This relationship forms the basis of CISK parameterization which is widely used in studying moist baroclinic processes (e.g., Shukla 1978; Mak 1982; O'Gorman 2011). The variables (temperature, zonal and meridional winds, and vertical velocity) for calculating diabatic heating rate as in Eq. (3) and training the regression model as in Eq. (4) are obtained from the fifth major global reanalysis produced by ECMWF (ERA5) (Hersbach et al. 2020), with a horizontal resolution of 0.25° × 0.25°. The regression model is trained using the springtime data from 2000 to 2010. The estimated coefficients are found not sensitive to changes in the period of the training data used.

Substituting Eq. (4) into Eq. (2), we obtain a QG thermodynamic equation with diabatic heating empirically incorporated:

$$\frac{\partial}{\partial t}\left[\frac{(Z_1 - Z_3)}{\delta p}\right] + \frac{1}{f\delta p}J(Z_3, Z_1) = \omega_2\left(S_2 + \frac{R\alpha}{C_p p_2}\right) = \omega_2\widehat{S}_2. \tag{5}$$

The impact of diabatic heating is equivalent to adding to S_2 in the dry model a "Q term" (the second term of $\widehat{S}_2$). This Q term is proportional to the local sensitivity of Q to ω_2 (i.e., α). The regression coefficient α is negative everywhere, reflecting that positive diabatic (latent) heating typically occurs with rising motion, thus negative ω. This approach mirrors the concept of "effective static stability" proposed in earlier studies (O'Gorman 2011; Booth et al. 2015). In this study, S_2 and α are, respectively, the stability parameter and regression coefficient defined *locally at individual model grid points*, which allows for the investigation of heating effects from selected regions in the model. In this study, we specifically examine the heating impact of the U.S. MCS in spring over a domain defined by 25°–55°N, 70°–105°W ("MCS region").

It is important to note that the highly idealized model used in this study does not resolve the intricate structures of MCS-associated heating such as the "top-heavy" profile, and the heating within the MCS region during springtime is not exclusively contributed by MCSs. However, within the MCS region defined above, MCSs indeed contribute dominantly to the latent heating in boreal spring. This assertion is supported by an MCS dataset developed in our prior research, where we quantified MCS activity with an automated detection algorithm. Our detection algorithm has been proven effective in reproducing some key MCS statistics over the Great Plains (You and Deng 2022). According to the MCS dataset, 47% of the hours during springtime 2000–20 are MCS active and heating during these hours contributes to 80%–100% of the total spring diabatic heating over the MCS region (not shown). We therefore term diabatic heating over the MCS region as "MCS heating" to highlight the important role played by these mesoscale systems in generating hemispheric-scale dynamical response.

Finally, combining Eqs. (1) and (5) by eliminating ω_2, we obtain the QGPV equation at the upper and lower levels used in nonmodal instability analysis:

$$\begin{aligned}\frac{\partial}{\partial t}\left[\zeta_1 + \frac{f(Z_3 - Z_1)}{(\delta p)^2\widehat{S}_2}\right] + f^{-1}J(Z_1, \zeta_1 + f) + \frac{1}{(\delta p)^2\widehat{S}_2}J(Z_1, Z_3) &= -\epsilon_1\zeta_1,\\ \frac{\partial}{\partial t}\left[\zeta_3 + \frac{f(Z_1 - Z_3)}{(\delta p)^2\widehat{S}_2}\right] + f^{-1}J(Z_3, \zeta_3 + f) + \frac{1}{(\delta p)^2\widehat{S}_2}J(Z_3, Z_1) &= -\epsilon_3\zeta_3.\end{aligned} \tag{6}$$

b. *Nonmodal instability (optimal mode) analysis*

The nonmodal instability analysis for this moist two-layer QG model is conducted over a channel domain bounded in the meridional direction by 20° and 65°N with a horizontal resolution of 2.5° × 2.5°, which is sufficient for resolving large-scale atmospheric disturbances. The regression coefficient α is interpolated to the model resolution from its original ERA5 grids, which vary spatially across the model domain. The purpose of the nonmodal instability (optimal mode) analysis is to identify the initial disturbances that amplify most rapidly over a predetermined time interval on a prescribed background flow within a dynamical system. The steps follow in general those discussed in Zhao et al. (2018, 2020) and Zhao and Deng (2020). These "optimally developing" initial disturbances (optimal modes or optimal disturbances) are constructed as a linear combination of all intrinsic normal modes (solved through an eigenvalue problem) of the model. The actual coefficients of the linear combinations are then obtained through an additional eigenvalue problem connecting directly with the "amplification factor." This step ultimately gives us the structures of the optimal modes for the preselected time interval and their corresponding amplification rate in the prescribed background flow.

First, we solve for the normal modes of the dynamical system formulated by Eq. (6), which is linearized about the observed spring mean flow. Assuming plane wave solutions, we write the upper- and lower-level perturbations as

$$Z_j'(\lambda, \varphi, t) = \Phi_j(\lambda, \varphi)e^{\sigma t}, \quad j = 1, 3, \tag{7}$$

where $\Phi(\lambda, \varphi)$ is the perturbation amplitude at a given model grid point (latitude φ and longitude λ). σ is the complex

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frequency with the real part indicating growth rate of the perturbation amplitude and the imaginary part indicating the perturbation frequency.

Substituting Eq. (7) into the linearized equations and applying finite differencing with cyclic boundary conditions, we obtain an equation in matrix form:

$$\sigma \mathbf{L}(\mathbf{\Phi}_1, \mathbf{\Phi}_3)^{\mathrm{T}} = \mathbf{R}(\mathbf{\Phi}_1, \mathbf{\Phi}_3)^{\mathrm{T}}. \tag{8}$$

Here, $\mathbf{L}$ and $\mathbf{R}$ are the square matrices of size $2N \times 2N$, where N is the number of grid points in the model horizontal domain. These matrices are linear operators on the left-hand side and right-hand side of the equations we obtained after applying wave solution [Eq. (7)] to the finite-difference format of the linear dynamical system. The matrices contain all the information about this dynamical system such as the predetermined parameters introduced in section 2a, as well as the climatological springtime geopotential, static stability parameters, and local values of α derived from ERA5. Equation (8) is solved as an eigenvalue problem where σ is the eigenvalue and $(\mathbf{\Phi}_1, \mathbf{\Phi}_3)^{\mathrm{T}}$ is the corresponding eigenvector, yielding $2N$ pairs of eigenvalues and eigenvectors. Each pair corresponds to one normal mode of the two-layer model.

We then use the obtained normal modes as base functions to derive optimal modes for a specified optimization time τ, which aims to identify an initial perturbation that intensifies the most over the time interval τ in the specified basic flow. The geopotential perturbation at $t = \tau$ can be expressed as

$$\boldsymbol{\psi} = (\boldsymbol{\psi}_1, \boldsymbol{\psi}_3) = \sum_{i=1}^{2N} (\mathbf{\Phi}_{1i}, \mathbf{\Phi}_{3i}) e^{\sigma_i \tau} a_i = \mathbf{P}\mathbf{\Lambda}\mathbf{a}. \tag{9}$$

Here, $\mathbf{P}$ is a $2N \times 2N$ matrix whose columns are composed of normal modes $(\mathbf{\Phi}_{1i}, \mathbf{\Phi}_{3i})$, $\mathbf{\Lambda}$ is a diagonal $2N \times 2N$ matrix with diagonal elements $\{e^{\sigma_i \tau}\}$, and $\mathbf{a}$ is a vector containing the projection coefficients $\{a_i\}$ of the initial perturbation onto the normal modes. Following Mak (2011), the intensity of a perturbation at $t = \tau$ is calculated as

$$\mathcal{A}(\tau) = \boldsymbol{\psi}^{\mathrm{H}} \mathbf{D} \boldsymbol{\psi} = \mathbf{a}^{\mathrm{H}} \mathbf{\Lambda}^{\mathrm{H}} \mathbf{P}^{\mathrm{H}} \mathbf{D} \mathbf{P} \mathbf{\Lambda} \mathbf{a} = \mathbf{a}^{\mathrm{H}} B(\tau) \mathbf{a}, \tag{10}$$

where $\boldsymbol{\psi}^{\mathrm{H}}$ is the Hermitian transpose of $\boldsymbol{\psi}$, and $\mathbf{D}$ is an identity matrix if Euclidean norm is used. This is the default setup for "global optimization" (i.e., optimization across the entire model domain) in which all grid points are considered to carry the same weights in measuring the perturbation intensity. If perturbation growth over a particular region is to be emphasized, the diagonal elements of $\mathbf{D}$ corresponding to this region can be kept as 1 while other diagonal elements are set to small values such as 0.01. We have adopted this "local optimization" to investigate the growth of synoptic-scale disturbances over the North America–North Atlantic sector (20°–65°N, 0°–130°W) in section 3c.

The amplification factor (the ratio between an optimal mode's amplitude at the end of the selected time interval and its initial amplitude) of a disturbance is defined as

$$\gamma = \frac{\mathbf{a}^{\mathrm{H}} B(\tau) \mathbf{a}}{\mathbf{a}^{\mathrm{H}} B(0) \mathbf{a}}. \tag{11}$$

Equation (11) can be further reduced to a new eigenvalue problem:

$$\gamma B(0)\mathbf{a} = B(\tau)\mathbf{a}. \tag{12}$$

This again leads to $2N$ pairs of eigenvalues γ and eigenvectors $\mathbf{a}$ and thus $2N$ optimal modes for every specified optimization time τ. For a given τ, the optimal mode with the largest amplification factor γ is denoted as the first optimal mode. The remaining optimal modes are ranked by decreasing values of γ. From Eq. (9), the spatial structure of each optimal mode (corresponding to one pair of γ and $\mathbf{a}$) at time t follows

$$\boldsymbol{\psi}(\lambda, \phi, t) = \sum_{i=1}^{2N} (\mathbf{\Phi}_{1i}, \mathbf{\Phi}_{3i}) e^{\sigma_i t} \mathbf{a}_i. \tag{13}$$

The spatiotemporal evolution of optimal disturbances can therefore be easily obtained.

3. Results

a. Empirical heating parameterization

To match the "local" nature of the heating parameterization described in section 2a, the horizontal variations of mid-level (500 mb) static stability S_2 are also introduced into the two-layer model as illustrated in Fig. 1a. Accounting for such variations in S_2 is necessary in this idealized analysis since the effect of diabatic heating is determined by the magnitude of the Q term relative to that of the local static stability. In the QG system, static stability across the channel domain is quantified using the formula

$$S_2 = \frac{R}{p}\left(\frac{RT}{pC_p} - \frac{\partial T}{\partial p}\right). \tag{14}$$

Values are smoothed near the Tibetan Plateau (25°–40°N, 75°–105°E; the blue rectangle in Fig. 1a) to mitigate the impacts of complex terrain (gray contours in Fig. 1). Higher values of S_2 are typically observed at lower latitudes compared to higher latitudes. When heating (Q term) is incorporated following Eq. (5), the resulting "effective" stability shows a substantially different spatial structure (Fig. 1c). Overall static stability decreases due to the negative correlation between Q and ω (i.e., negative values of α). This is consistent with the understanding that positive latent heating associated with saturated ascent is the main component of the diabatic heating. The most pronounced reductions are consistent with regions experiencing the most intense diabatic heating, such as the North Pacific storm-track region, the North Atlantic storm-track region, and the MCS-active region over the central United States (Fig. 1b). Figure 1d shows the distribution of R^2 values of the local regression across the model domain. In the three major diabatic heating regions, the regression model accounts for approximately 70% of variance. This underscores

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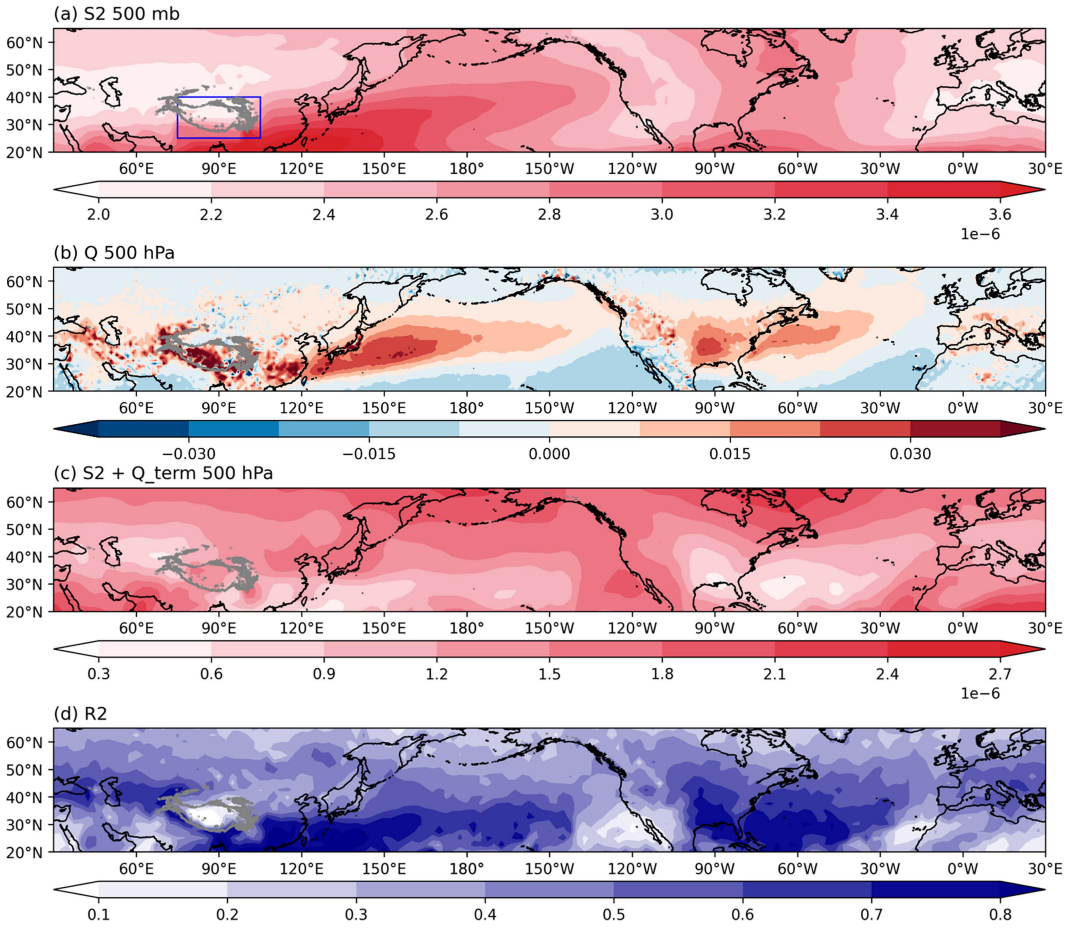


FIG. 1. Spatial distribution of (a) mean stability parameter at 500 mb (color shading; $m^2\ s^{-2}\ Pa^{-2}$) of springtime 2000–20, (b) mean diabatic heating rate at 500 mb (Q; $J\ kg^{-1}\ s^{-1}$), (c) stability parameter adjusted by adding the diabatic heating term, and (d) R^2 values of the regression between local diabatic heating rate and vertical velocity. Gray contours in each panel denote an altitude of 4500 m. The blue rectangle in (a) indicates the area where the stability parameter was smoothed.

the effectiveness of our parameterization in reproducing the diabatic heating in these major heating areas.

Figure 2 presents scatterplots of Q versus ω at three selected points within each of the three major heating regions. They all demonstrate clear negative correlations between Q and ω. It is worth noting that there is inherent dynamical nonlinearity in the relationship between latent heating and vertical velocity because condensation and freezing are mostly associated with ascent, whereas in sinking motion, the contribution from radiative heating increases. This nonlinearity presents a challenge in any linear parameterization of Q based on ω, as demonstrated in Fig. 2, where the linear regression tends to slightly underestimate heating during ascent and overestimate cooling during descent. These limitations, however, do not qualitatively affect the results of the linear instability analysis here due to the predominance of latent heating in those major heating regions (not shown).

The empirical heating parameterization used in this study is further validated through individual weather events characterized by extreme precipitation and a significant amount of latent heating. One such event analyzed is the extratropical cyclone over the United States on 20–21 May 2011. This cyclone is accompanied by a well-documented MCS event during the Midlatitude Continental Convective Clouds Experiment (MC3E) field campaign (Jensen et al. 2016; Marinescu et al. 2016; Fan et al. 2017; Han et al. 2019; Varble et al. 2020). Based on data from ERA5, Figs. 3a and 3b show an intense cyclone centered over the central United States, with stratiform precipitation on the northeastern side and associated diabatic heating [calculated based on Eq. (3)] at 500 mb along the frontal boundary. At 0000 UTC, a substantial heating zone associated with the MCS developed over Texas with a spatial scale of hundreds of kilometers. By 0600 UTC, this convective zone split into two, and significant cooling developed on the rear side of the MCS where the rear inflow jet led to strong evaporation of condensates, a phenomenon often seen in mature MCSs (e.g., Houze 2004). The regression model's prediction (Figs. 3c,d) reproduces the overall structure of diabatic heating around the cyclone,

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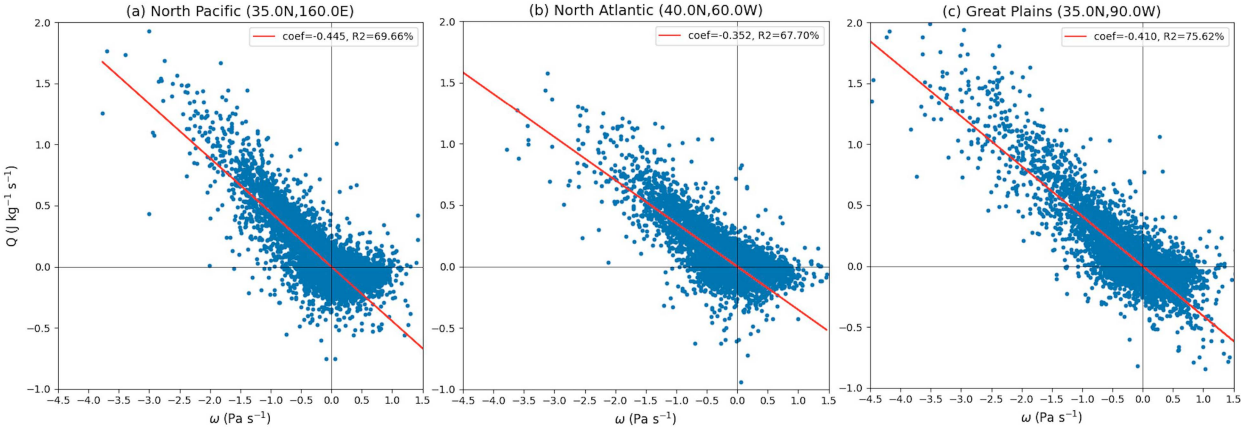


FIG. 2. Scatterplots showing the relationship between diabatic heating rate (Q; J kg^{-1} s^{-1}) and vertical velocity (ω; Pa s^{-1}) at three selected locations: over (a) North Pacific (35°N, 160°E), (b) North Atlantic (40°N, 60°W), and (c) Great Plains (35°N, 90°W). Each plot includes a linear regression line (red) with corresponding regression coefficients and R^2 values.

despite a tendency to underestimate heating in stratiform areas and slightly overestimate heating and cooling in areas following the MCS event over Texas. We have evaluated many other weather events (not shown) to ensure that the parameterization is well founded and is sufficient for the purpose of our study. In general, the model can accurately replicate the heating structures across all tested events, which demonstrates the reliability of the linear parameterization.

b. *Impact of domainwide heating on the synoptic variability over the northern midlatitudes*

We begin our analysis by assessing the impact of domainwide diabatic heating (across the entire channel domain 20°–65°N) on the development of synoptic-scale disturbances over the northern midlatitudes. The observed springtime synoptic variability is shown in Fig. 4, which illustrates the top four empirical orthogonal functions (EOFs) of bandpass-filtered (2–8 days)

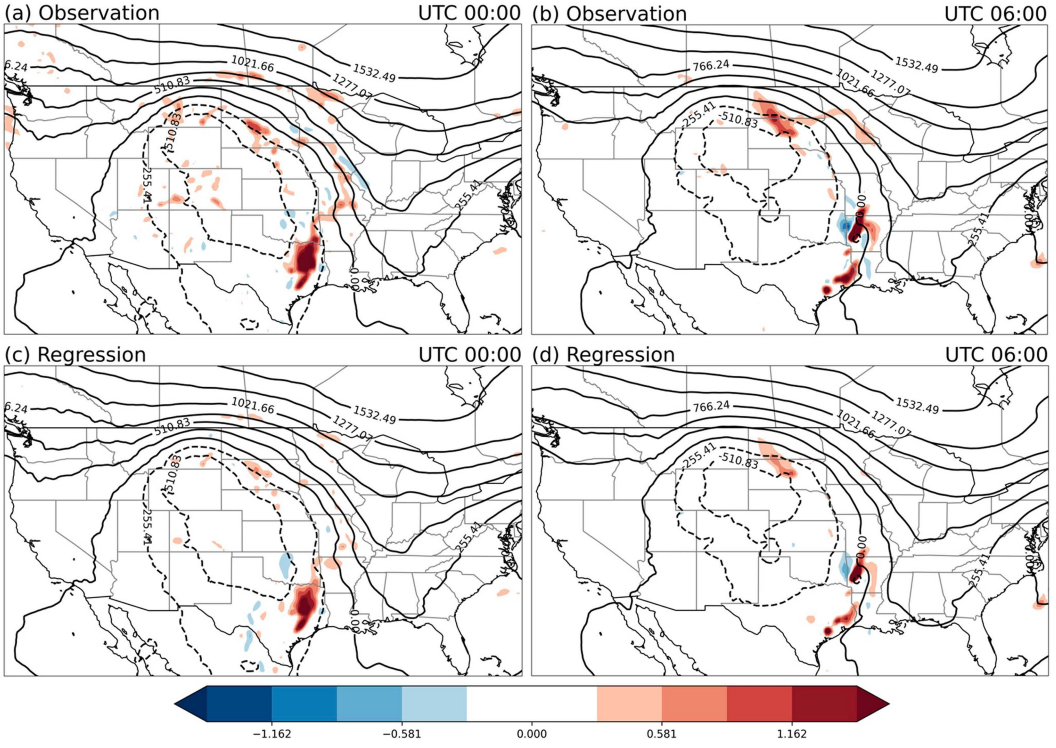


FIG. 3. Diabatic heating rate (J kg^{-1} s^{-1}) and 750-mb geopotential anomalies (m^2 s^{-2}) at 0000 UTC 21 May 2011 and 0600 UTC 21 May 2011. Color shading in (a) and (b) represents diabatic heating rates derived from ERA5 data, (c), (d) estimates generated by the regression model. Contours illustrate 750-mb geopotential anomalies across all panels.

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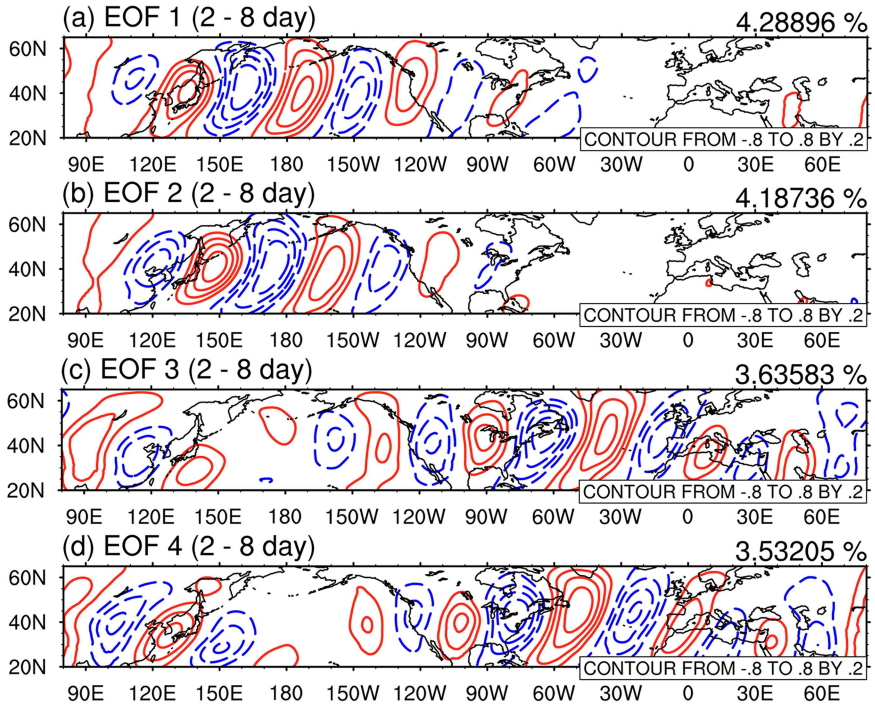


FIG. 4. Top four EOF modes from ERA5 bandpass-filtered (2–8 days) 250-mb geopotential anomalies across the channel domain (20°–65°N) during spring from 2000 to 2020.

250-mb geopotential during springtime 2000–20. The top two modes capture disturbances originating from the North Pacific storm-track eddies and propagating downstream toward North America. These disturbances possess a zonal wavelength of approximately 4000 km, reflecting typical synoptic scales in spring. The third and fourth EOFs feature synoptic disturbances with maximum amplitudes over and near North America, traveling across the North Atlantic, the Mediterranean, and central Asia and eventually reentering the western Pacific. Collectively, the Pacific EOF modes account for about 8.5% of the total variance, whereas the Atlantic–Eurasia EOF modes contribute about 7.2%. These EOF modes are representative of the springtime synoptic variability over the northern midlatitudes.

To examine how well the two-layer QG model captures these observed features of synoptic variability, we examine the optimal modes corresponding to optimization times of 1, 2, 3, and 4 days, which are all relevant for the development of synoptic-scale disturbances. The spatiotemporal evolution of these optimal modes indeed demonstrates features of synoptic-scale disturbances as in Fig. 4. One example is presented in Fig. 5, which depicts the evolution of the first optimal mode with $\tau = 3$ days from day 0 to 9 under both the dry [no heating; Figs. 5a(1)–a(3)] and moist [with heating everywhere over the model domain; Figs. 5b(1)–b(3)] scenarios. In both scenarios, the disturbances are initiated over the East Asian coast and the western Pacific and mature by day 3. These disturbances with southeast–northwest (SE–NW) tilts lean along the background shears poleward of the seasonal-mean subtropical jet, suggesting a barotropic conversion of kinetic energy from the disturbances to the background flow. Therefore, under both the dry and moist settings, baroclinic conversion is mainly responsible for the growth of the disturbances, characterized by the classic "westward tilt with height" indicated by the upper- (contours) and lower- (color shading) level disturbances in Fig. 5. Diabatic heating more than doubles the amplification factor of the first optimal mode, which increases from 84.6 in the dry case to 177 in the moist case (left vs right columns of Fig. 5). This change is comparable with results found in earlier studies (e.g., Booth et al. 2015).

Building on the assessment of individual optimal modes, we further explore the overall synoptic variability in the two-layer model and compare with observations (Fig. 4). Specifically, we construct a hypothetical flow consisting of day 1–15 evolutions of individual optimal modes randomly sampled from the top 50 optimal modes with $\tau = 1$, 2, 3, and 4 days (200 modes in total). The sampling (with replacement) is repeated 500 times to generate a flow field spanning 7500 days (15 days $\times$ 500 samples). The top EOFs of the geopotential disturbances are obtained from this hypothetical flow for the upper level (250 mb) and are shown in Fig. 6. In the dry model (Figs. 6a,b), EOF 1 features wave trains with peak amplitudes over the Eurasian continent, different from those in observation where the disturbance amplitude peaks over the western North Pacific (Figs. 4a,b). EOF 1 is also characterized by larger zonal scales compared to those observed. EOF 3, meanwhile, resembles synoptic-scale disturbances over the North America–North Atlantic sector, with horizontal scales close to those observed (Figs. 4c,d). EOF modes 2 and 4 are also excluded from the figure as they reflect, respectively, the eastward propagation of EOFs 1 and 3.

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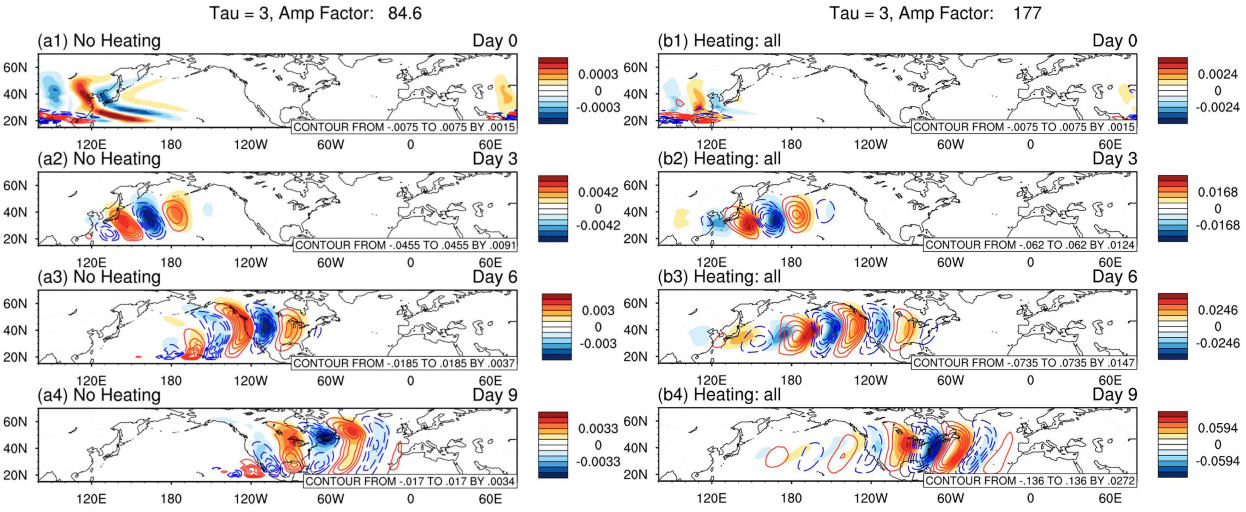


FIG. 5. Evolution of the first optimal mode with $\tau = 3$ days, from day 0 to 9, comparing scenarios of [a(1)]–[a(4)] no heating and [b(1)]–[b(4)] with heating everywhere across the model domain. The color shading represents disturbances at the lower level (750 mb), while the contours represent disturbances at the upper level (250 mb).

When diabatic heating is activated over the entire model domain, the structures of the EOFs change substantially (Figs. 6c,d). Compared to the dry case, EOF 1 in the moist model indicates better-defined synoptic disturbances, reaching maximum amplitude over the North Pacific with enhanced meridional extent and reduced zonal scales (Fig. 6c vs Fig. 6a). EOF 1 in the moist model thus aligns more closely with EOFs 1 and 2 derived from the ERA5 reanalysis (Figs. 4a,b). Similarly, EOF 3 (Fig. 6d) in the moist case also reflects better the characteristics of EOFs 3 and 4 in Figs. 4c and 4d, featuring greater

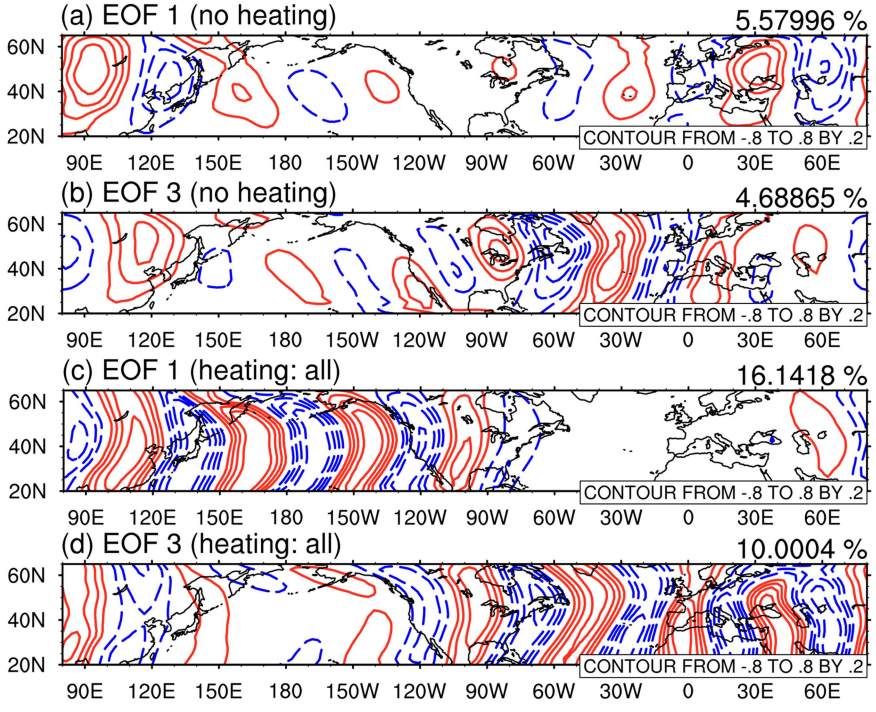


FIG. 6. EOF 1 and EOF 3 of the upper-level (250 mb) hypothetical flows from the two-layer model, under two heating scenarios: (a),(b) no heating and (c),(d) complete heating across the model domain. The hypothetical flows are constructed by randomly sampling 500 times with replacement, with each sample covering a 15-day evolution from the top 50 optimal modes of $\tau = 1, 2, 3$, and 4 days. The construction method is further detailed in the main text.

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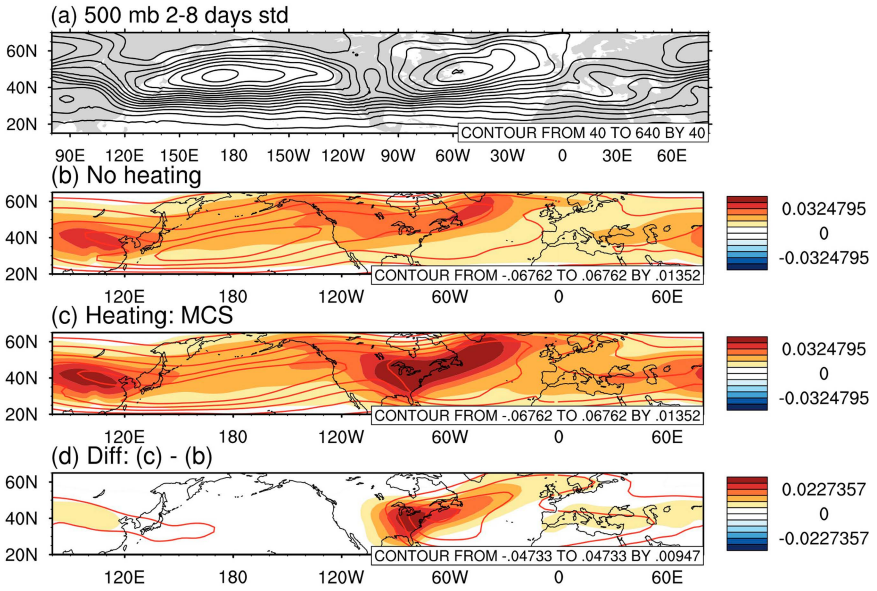


FIG. 7. (a) Standard deviation of 2–8-day bandpass-filtered 500-mb geopotential during springtime 2000–20. (b)–(d) The standard deviation of hypothetical flows constructed from sampled channel optimal modes, under different heating scenarios: (b) no heating, (c) MCS heating only, and (d) the difference between scenarios (b) and (c). Color shading represents standard deviation at the lower level, while contours correspond to the higher level.

meridional extents and more pronounced downstream development of disturbances over the North Atlantic and Europe. The variance contributions of the top four EOFs increase significantly from the dry to the moist model, even though the exact values cannot be compared to those based on observation due to the highly idealized nature of the hypothetical flow constructed. Therefore, incorporating flow-dependent diabatic heating in the QG model improves the model's capability of reproducing the general characteristics (scales, spatial distributions, etc.) of the synoptic variability in the northern midlatitudes in spring.

c. *Footprints of the U.S. MCS heating in the synoptic variability of the northern midlatitudes*

We next focus on the impact of regional U.S. MCS heating (heating in MCS region 25°–55°N, 70°–105°W, defined in section 2a) on the hemispheric synoptic variability across the northern midlatitudes. To isolate the effect of diabatic heating originating from the MCS region, we conduct the optimal mode analysis for τ = 1, 2, 3, and 4 days under four distinct heating scenarios: 1) completely dry (no heating), 2) MCS heating only (MCS), 3) heating activated across the domain except for the MCS region (all MCS), and 4) domainwide heating (all). Conducting such a comprehensive analysis allows us to contrast the role of MCS heating with the effect of heating outside of the MCS region. Figures 7b–d display the standard deviation of the disturbances under heating scenarios 1 and 2 based upon hypothetical flows constructed following the method described in section 3b. In the completely dry scenario (Fig. 7b), the model fails to capture the North Atlantic storm track at the upper level (contours) compared to the observation (Fig. 7a). The lower-level Pacific storm track is also found over the East Asian continent instead of the North Pacific (color shading). The introduction of the MCS heating significantly enhances synoptic variability at both upper and lower levels over the North American–North Atlantic sector, effectively reproducing some features of the North Atlantic storm track (Fig. 7c). The changes induced by MCS heating extend downstream reaching as far as the central and East Asia (Fig. 7d) despite the absence of diabatic heating outside the MCS region.

Figure 8 shows the standard deviation of the sampled optimal disturbances under two other heating scenarios: "all MCS" and "all," along with their differences. When heating is applied across the entire model domain (Fig. 8b), the model successfully reproduces the two storm tracks despite the slight westward and equatorward shifts of their cores because of the idealized and linear nature of the analysis. This finding is consistent with earlier EOF results (Figs. 6c,d), demonstrating the important role of domainwide diabatic heating (mostly heating outside of the MCS region) in establishing the gross features of the northern midlatitude synoptic variability. When the U.S. MCS heating is excluded from the model (Fig. 8a), the North Atlantic storm track nearly vanishes, reaffirming the signals seen in Fig. 7. The footprints of MCS heating become more prominent when heating is also activated everywhere else: Fig. 8c indicates a more significant boost of synoptic variability over a greater zonal extent and intensity compared to Fig. 7d. In summary, the diabatic (latent) heating associated with MCS development over the United States not only intensifies local synoptic activity as expected from theories of moist baroclinic growth but also exerts a substantial influence on its downstream

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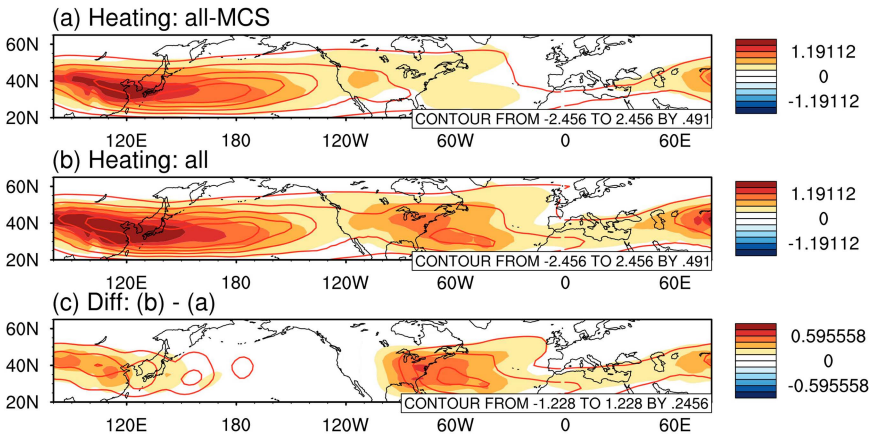


FIG. 8. As in Figs. 7b–d, but for heating scenarios: (a) all heating activated except for MCS region, (b) domainwide heating activated, and (c) the difference between (b) and (a).

regions, extending its impact to planetary and hemispheric scales. The addition of heating outside of the MCS region, particularly over the North Atlantic, enhances the impact of MCS heating on the downstream synoptic variability.

d. *Footprints of MCS heating seen in "local optimal modes"*

Given the fact that MCS heating impacts peak over the North American–North Atlantic sector, we further delineate these impacts in the context of local optimal modes–optimal disturbances obtained through optimization over this sector (20°–65°N, 0°–130°W) (see the discussion in section 2b). Figure 9 illustrates the evolution of the first local optimal mode from day 3 to 7 for $\tau = 3$ days under four distinct heating scenarios: no heating [Figs. 9a(1)–a(3)], MCS only [Figs. 9b(1)–b(3)], all heating except for the MCS region (all MCS) [Figs. 9c(1)–c(3)], and domainwide heating (all) [Figs. 9d(1)–d(3)]. Without heating [Figs. 9a(1)–a(3)], by day 3, the disturbance reaches western North America. It peaks around day 5 with a pronounced trough-ridge pattern influencing North America and eventually decays over the North Atlantic. With only MCS heating activated [Figs. 9b(1)–b(3)], the disturbance follows a similar structure as the dry scenario, but the amplitude of the disturbance is amplified, especially over the North Atlantic region. In the all-MCS scenario [Figs. 9c(1)–c(3)], there is a noticeable increase in the meridional extent of the local optimal disturbances as well as their amplitude over the North Atlantic when compared to the no-heating scenario. Further addition of the MCS heating [Figs. 9d(1)–d(3)] leads to more intense disturbances on days 5 and 7, confirming earlier findings of MCS heating footprints based upon "global" instead of local optimal modes.

Building on the approach described in section 3c, we construct hypothetical flows using sampled local optimal modes with $\tau = 1, 2, 3$, and 4 days to evaluate the model's capability of reproducing the synoptic variability across the North America–North Atlantic sector. Figure 10a presents the top EOF of the 2–8-day bandpass-filtered 250-mb geopotential over the North American–North Atlantic sector based on the ERA5 reanalysis, which is characterized by large-scale disturbances with horizontal scales close to EOFs 3 and 4 when the EOF analysis was applied to the entire channel domain (i.e., Figs. 4c,d). Figures 10b–e then show the top EOFs obtained from hypothetical flow fields generated through random sampling of the top local optimal modes under the four heating scenarios. Consistent with the individual optimal mode presented in Fig. 9, adding only MCS heating to the dry model does not significantly alter the optimal mode structure, such as the horizontal scales of the dry local optimal modes (Fig. 10c vs Fig. 10b). In contrast, incorporating heating over all other regions except for MCS significantly reduces the predominant zonal scales of the optimal modes, aligning them more closely with observed synoptic disturbances (Fig. 10d vs Fig. 10b). Further addition of the MCS heating mostly increases the amplitude of the synoptic variability in the model with the structure of disturbances largely unchanged (Fig. 10d vs Fig. 10e). In the domainwide heating scenario, the maximum spatial correlation between the local optimal modes and the top observed mode of synoptic variability (Fig. 10a) reaches 0.87 and an average of 0.57 for the top 10 optimal modes corresponding to $\tau = 1, 2, 3$, and 4 days. This is a drastic improvement from the maximum correlation of 0.62 and an average correlation of 0.37 in the dry model.

Although MCS heating alone does not produce a decisive impact on the horizontal structure of the local optimal modes and synoptic variability in the model, it does enhance the amplitude of the unstable disturbances, especially during its later stage of the development (e.g., days 5 and 7 in Figs. 9c,d). This suggests that synoptic-scale disturbances in the North Atlantic sector may saturate, enter the nonlinear stage, and break earlier as a result of the MCS heating over the United States, holding significant implications for the development of low-frequency variability in this region. Following Hoskins et al. (1983), we examine the transient (synoptic) eddy forcing on the background zonal flow. Specifically, we compute the horizontal component of the upper-tropospheric **E** vector:

$$\mathbf{E} = (\overline{v'^2 - u'^2},\ -\overline{u'v'}), \tag{15}$$

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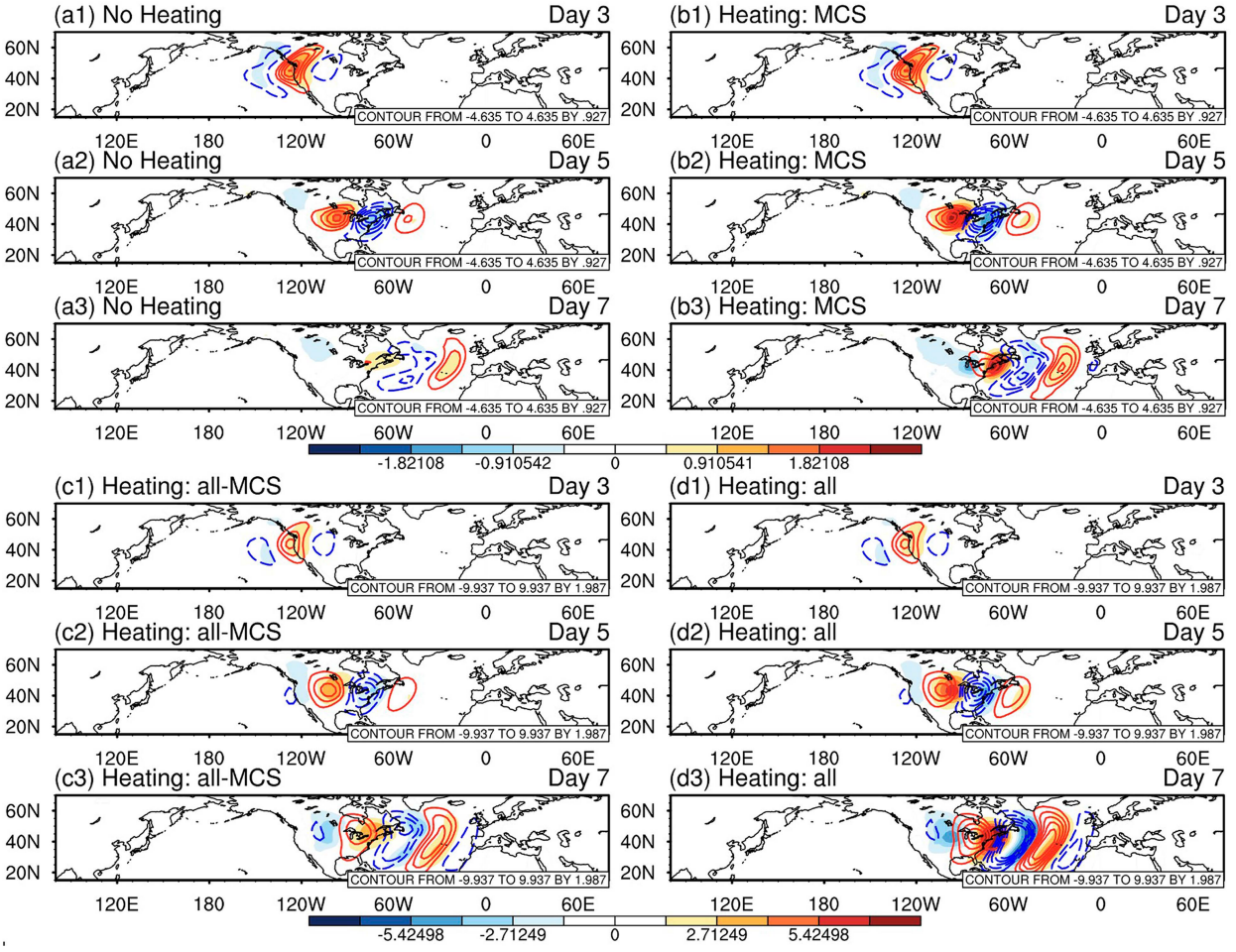


FIG. 9. Evolution of the first local optimal mode with $\tau = 3$ days, from day 3 to 7, under four different heating scenarios: [a(1)]–[a(3)] without any diabatic heating, [b(1)]–[b(3)] with diabatic heating specifically in the MCS region, [c(1)]–[c(3)] all heating except for the MCS region, and [d(1)]–[d(3)] domainwide heating. The color shading illustrates disturbances at the lower level (750 mb), while the contours delineate disturbances at the upper level (250 mb).

where u' and v' are zonal and meridional geostrophic winds, respectively, calculated from geopotential height anomalies of the sampled hypothetical flows. The divergence (convergence) of **E** vector corresponds to an acceleration (deceleration) of the local background zonal wind. Figure 11 illustrates the divergence of **E** vector averaged over the hypothetical flows under various heating scenarios. In the dry scenario (no heating), significant divergence of **E** vector is found downstream and slightly northward of the background North Atlantic jet at approximately 50°N, reflecting a general tendency for model disturbances to maintain and extend the time mean jet downstream through barotropic processes. This result is also consistent with the general southwest–northeast (SW-NE) tilts of optimal modes that imply poleward fluxes of eddy momentum (Figs. 10b,c). The magnitude of **E** vector divergence and convergence and thus the eddy forcing of the mean zonal wind nearly double with the inclusion of MCS heating [Figs. 11a(2),a(3)] and indicate a tendency of model disturbances to extend and bifurcate the background jet. When heating in other regions is factored in Figs. 11b(1)–b(3), the impact of MCS heating on eddy forcing becomes much stronger and more concentrated [Figs. 11b(2),b(3)]. The eddy forcing over the North Atlantic features a meridional dipole structure indicating a poleward flux of zonal momentum by eddies. The center of the mean zonal wind acceleration (divergence) is concentrated slightly southward of the background jet. In summary, MCS heating leads to more intense synoptic disturbances and stronger eddy forcing on the background zonal wind, which strengthens the Atlantic jet and extends it downstream. The impact of MCS heating on eddy forcing also depends on heating at other regions, which enhances the extension and southward shift of the background jet when heating in the other regions is turned on in the model.

4. Concluding remarks

In a two-layer QG channel model with empirical parameterizations of diabatic heating, this study employs nonmodal

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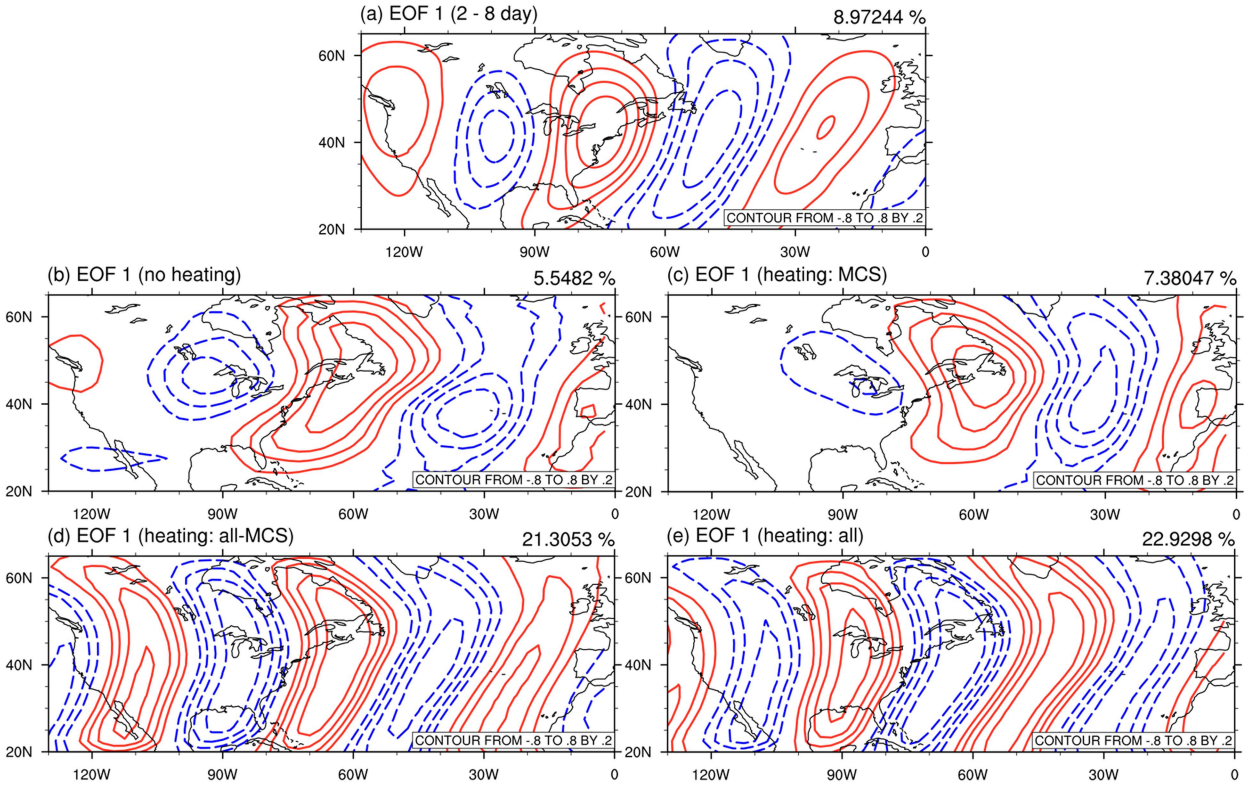


FIG. 10. (a) The first EOF of ERA5 bandpass-filtered (2–8 days) 250-mb geopotential over the North America–North Atlantic sector (20°–65°N, 0°–130°W), during springtime 2000–20. (b)–(e) The first EOF over the North Atlantic sector of 250-mb hypothetical flows constructed by randomly sampling local optimal modes, under four heating scenarios: (b) no heating, (c) MCS heating only, (d) all heating except for the MCS region, and (e) domainwide heating included.

instability analysis to detect and characterize the footprints of the U.S. MCS heating in the springtime synoptic-scale circulation of the northern midlatitudes. The heating parameterization, in terms of local linear regression onto the vertical velocity at the model midlevel, successfully reproduces the overall intensities and spatial characteristics of flow-dependent diabatic heating. Four heating scenarios are considered in the investigation of the moist optimal modes (disturbances) developing in the observed climatological spring background flow. MCS heating over the United States is found to be critical in creating at the model upper level a second maximum of synoptic variability, i.e., the North Atlantic storm track, when the optimization is done over the entire model domain. Including diabatic heating in this highly idealized model in general reduces the zonal scale and increases the meridional extent of optimal disturbances, leading to modes of synoptic variability (revealed via EOFs) that much more closely resemble the observed ones. When seeking maximum disturbance growth over a period of several days over the North America–North Atlantic sector through local optimization, it is found that domainwide heating excluding MCS heating, particularly the North Pacific and North Atlantic heating, sets the zonal scale of the disturbances. The further addition of MCS heating significantly increases the amplitude of optimal disturbances and model synoptic variability downstream of North America. Such intensified disturbances over the North Atlantic also exert stronger transient (synoptic) eddy forcing of the background zonal flow, leading to an overall strengthening and downstream extension of the background Atlantic jet.

In summary, diabatic heating (largely latent heating) is crucial for the development of the main features in the springtime synoptic variability over the northern midlatitudes. The footprints of MCS heating over the United States include 1) better separation of the North Atlantic storm track from the North Pacific storm track, 2) larger amplitudes of synoptic variability over the eastern North America and North Atlantic, 3) better organized circumglobal propagation of synoptic disturbances in the midlatitude channel, and 4) stronger eddy forcing of the background Atlantic jet. The effect of MCS heating is therefore of planetary- and hemispheric-scale and entangled with diabatic effects over other regions such as East Asia and North Pacific. The intensification by MCS heating of the synoptic disturbances over the North Atlantic suggests that the heating footprints may also be found in the modes of low-frequency variability (e.g., the northern annular mode), as the dynamics of these modes are closely linked to the nonlinear development and eventual breaking of synoptic waves (e.g., Benedict et al. 2004; Woollings et al. 2008;

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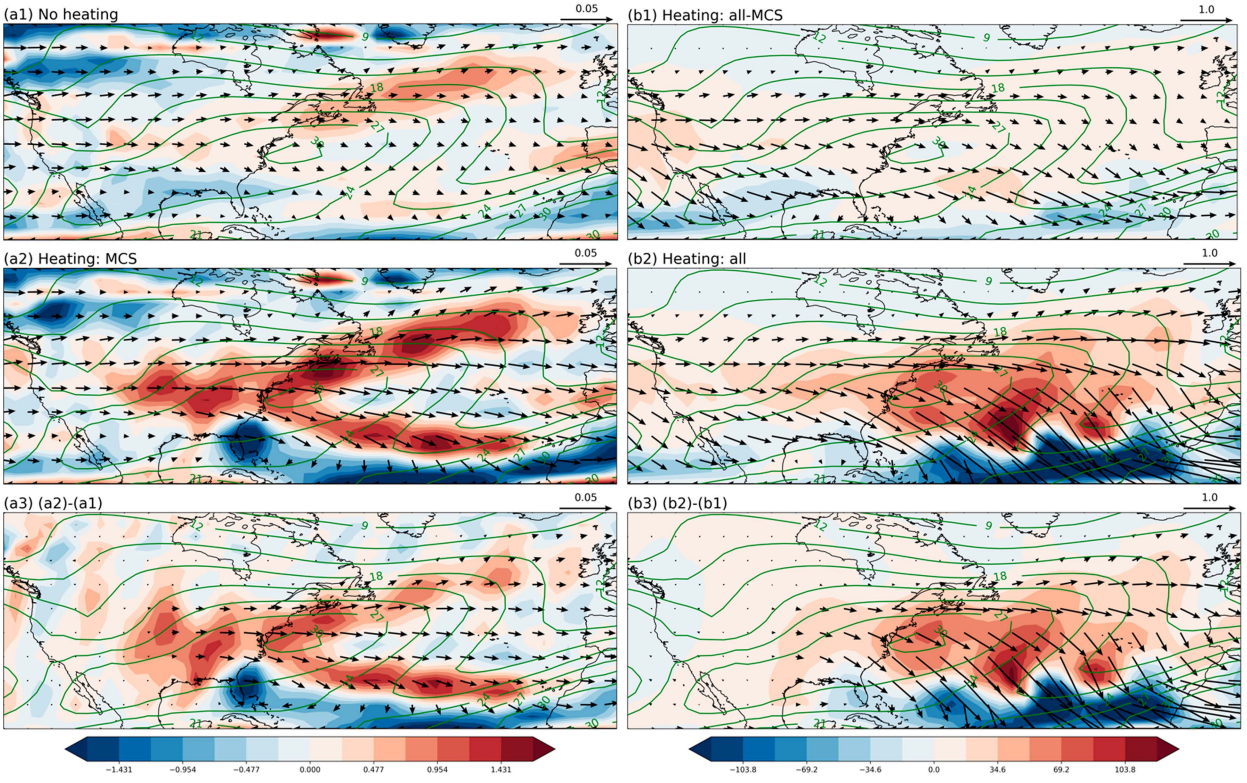


FIG. 11. The **E** vector (vector; 10^{-4} m^2 s^{-2}) and its divergence (color shading; 10^{-12} m s^{-2}) at 250 mb averaged over the hypothetical flows sampled from the local optimal modes under different heating scenarios: [a(1)] no heating, [a(2)] MCS heating included only, [b(1)] all heating included except for the MCS region, and [b(2)] domainwide heating included. [a(3)],[b(3)] The difference between [a(2)] and [a(1)] and [b(2)] and [b(1)], respectively. Contours throughout the panels represent the mean zonal wind (m s^{-1}) at 250 mb during springtime 2000–20.

Rivière and Orlanski 2007). The results reported here provide a basic context for understanding and attributing potential changes in the large-scale circulation and regional climate of the northern extratropics, given the anticipated increase in MCS frequency over the United States during the warm season (Prein et al. 2017). Due to the highly simplified and idealized nature of the model used here, many factors including the vertical profiles of convective heating and the separation of latent and radiative heating are left out of our analysis. These factors, together with the effect of MCS heating on the nonlinear life cycle of synoptic disturbances, remain to be investigated with models of increased complexity.

Acknowledgments. The authors thank three anonymous reviewers for their constructive comments that led to significant improvement of the paper. This study is support by the U.S. National Science Foundation (NSF) through Grant AGS-2032532 and by the U.S. National Oceanic and Atmospheric Administration (NOAA) through Grant NA22OAR4310606.

Data availability statement. The atmospheric variables on the single level and all pressure levels used in this study come from the fifth major global reanalysis produced by ECMWF (ERA5) and are available on the Climate Data Store (https://cds.climate.copernicus.eu/#!/home).

REFERENCES

Bartels, D. L., and R. A. Maddox, 1991: Midlevel cyclonic vortices generated by mesoscale convective systems. *Mon. Wea. Rev.*, **119**, 104–118, https://doi.org/10.1175/1520-0493(1991)119<0104:MCVGBM>2.0.CO;2.

Benedict, J. J., S. Lee, and S. B. Feldstein, 2004: Synoptic view of the North Atlantic Oscillation. *J. Atmos. Sci.*, **61**, 121–144, https://doi.org/10.1175/1520-0469(2004)061<0121:SVOTNA>2.0.CO;2.

Bentley, M. L., and J. A. Sparks, 2003: A 15 yr climatology of derecho-producing mesoscale convective systems over the central and eastern United States. *Climate Res.*, **24**, 129–139, https://doi.org/10.3354/cr024129.

Booth, J. F., L. Polvani, P. A. O'Gorman, and S. Wang, 2015: Effective stability in a moist baroclinic wave. *Atmos. Sci. Lett.*, **16**, 56–62, https://doi.org/10.1002/asl2.520.

Charney, J. G., and A. Eliassen, 1964: On the growth of the hurricane depression. *J. Atmos. Sci.*, **21**, 68–75, https://doi.org/10.1175/1520-0469(1964)021%3C0068:OTGOTH%3E2.0.CO;2.

Coniglio, M. C., J. Y. Hwang, and D. J. Stensrud, 2010: Environmental factors in the upscale growth and longevity of MCSs

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derived from rapid update cycle analyses. *Mon. Wea. Rev.*, **138**, 3514–3539, https://doi.org/10.1175/2010MWR3233.1.

Cui, W., T. J. Galarneau Jr., and K. A. Hoogewind, 2024: Changes in mesoscale convective system precipitation structures in response to a warming climate. *J. Geophys. Res. Atmos.*, **129**, e2023JD039920, https://doi.org/10.1029/2023JD039920.

Emanuel, K. A., 1994: *Atmospheric Convection.* Oxford University Press, 580 pp.

——, M. Fantini, and A. J. Thorpe, 1987: Baroclinic instability in an environment of small stability to slantwise moist convection. Part I: Two-dimensional models. *J. Atmos. Sci.*, **44**, 1559–1573, https://doi.org/10.1175/1520-0469(1987)044%3C1559:BIIAEO%3E2.0.CO;2.

Fan, J., and Coauthors, 2017: Cloud-resolving model intercomparison of an MC3E squall line case: Part I—Convective updrafts. *J. Geophys. Res. Atmos.*, **122**, 9351–9378, https://doi.org/10.1002/2017JD026622.

Feng, Z., L. R. Leung, S. Hagos, R. A. Houze, C. D. Burleyson, and K. Balaguru, 2016: More frequent intense and long-lived storms dominate the springtime trend in central US rainfall. *Nat. Commun.*, **7**, 13429, https://doi.org/10.1038/ncomms13429.

——, R. A. Houze Jr., L. R. Leung, F. Song, J. C. Hardin, J. Wang, W. I. Gustafson, and C. R. Homeyer, 2019: Spatiotemporal characteristics and large-scale environments of mesoscale convective systems east of the Rocky Mountains. *J. Climate*, **32**, 7303–7328, https://doi.org/10.1175/JCLI-D-19-0137.1.

Han, B., and Coauthors, 2019: Cloud-resolving model intercomparison of an MC3E squall line case: Part II. Stratiform precipitation properties. *J. Geophys. Res. Atmos.*, **124**, 1090–1117, https://doi.org/10.1029/2018JD029596.

Hersbach, H., and Coauthors, 2020: The ERA5 global reanalysis. *Quart. J. Roy. Meteor. Soc.*, **146**, 1999–2049, https://doi.org/10.1002/qj.3803.

Hoskins, B. J., I. N. James, and G. H. White, 1983: The shape, propagation and mean-flow interaction of large-scale weather systems. *J. Atmos. Sci.*, **40**, 1595–1612, https://doi.org/10.1175/1520-0469(1983)040<1595:TSPAMF>2.0.CO;2.

Houze, R. A., Jr., 2004: Mesoscale convective systems. *Rev. Geophys.*, **42**, RG4003, https://doi.org/10.1029/2004RG000150.

——, B. F. Smull, and P. Dodge, 1990: Mesoscale Organization of springtime rainstorms in Oklahoma. *Mon. Wea. Rev.*, **118**, 613–654, https://doi.org/10.1175/1520-0493(1990)118<0613:MOOSRI>2.0.CO;2.

Jensen, M. P., and Coauthors, 2016: The Midlatitude Continental Convective Clouds Experiment (MC3E). *Amer. Meteor. Soc.*, **97**, 1667–1686, https://doi.org/10.1175/BAMS-D-14-00228.1.

Lapeyre, G., and I. M. Held, 2004: The role of moisture in the dynamics and energetics of turbulent baroclinic eddies. *J. Atmos. Sci.*, **61**, 1693–1710, https://doi.org/10.1175/1520-0469(2004)061<1693:TROMIT>2.0.CO;2.

Mak, M., 1982: On moist quasi-geostrophic baroclinic instability. *J. Atmos. Sci.*, **39**, 2028–2037, https://doi.org/10.1175/1520-0469(1982)039%3C2028:OMQGBI%3E2.0.CO;2.

——, 2011: *Atmospheric Dynamics.* Cambridge University Press, 500 pp.

Marinescu, P. J., S. C. van den Heever, S. M. Saleeby, and S. M. Kreidenweis, 2016: The microphysical contributions to and evolution of latent heating profiles in two MC3E MCSs. *J. Geophys. Res. Atmos.*, **121**, 7913–7935, https://doi.org/10.1002/2016JD024762.

Montgomery, M. T., and B. F. Farrell, 1991: Moist surface frontogenesis associated with interior potential vorticity anomalies in a semigeostrophic model. *J. Atmos. Sci.*, **48**, 343–368, https://doi.org/10.1175/1520-0469(1991)048%3C0343:MSFAWI%3E2.0.CO;2.

Nigam, S., 1994: On the dynamical basis for the Asian summer monsoon rainfall-El Niño relationship. *J. Climate*, **7**, 1750–1771, https://doi.org/10.1175/1520-0442(1994)007%3C1750:OTDBFT%3E2.0.CO;2.

O'Gorman, P. A., 2011: The effective static stability experienced by eddies in a moist atmosphere. *J. Atmos. Sci.*, **68**, 75–90, https://doi.org/10.1175/2010JAS3537.1.

Prein, A. F., C. Liu, K. Ikeda, S. B. Trier, R. M. Rasmussen, G. J. Holland, and M. P. Clark, 2017: Increased rainfall volume from future convective storms in the US. *Nat. Climate Change*, **7**, 880–884, https://doi.org/10.1038/s41558-017-0007-7.

Rivière, G., and I. Orlanski, 2007: Characteristics of the Atlantic storm-track eddy activity and its relation with the North Atlantic Oscillation. *J. Atmos. Sci.*, **64**, 241–266, https://doi.org/10.1175/JAS3850.1.

Schumacher, R. S., and R. H. Johnson, 2006: Characteristics of U.S. extreme rain events during 1999–2003. *Wea. Forecasting*, **21**, 69–85, https://doi.org/10.1175/WAF900.1.

——, A. J. Clark, M. Xue, and F. Kong, 2013: Factors influencing the development and maintenance of nocturnal heavy-rain-producing convective systems in a storm-scale ensemble. *Mon. Wea. Rev.*, **141**, 2778–2801, https://doi.org/10.1175/MWR-D-12-00239.1.

Shukla, J., 1978: CISK-barotropic-baroclinic instability and the growth of monsoon depressions. *J. Atmos. Sci.*, **35**, 495–508, https://doi.org/10.1175/1520-0469(1978)035%3C0495:CBBIAT%3E2.0.CO;2.

Smith, B. T., R. L. Thompson, J. S. Grams, C. Broyles, and H. E. Brooks, 2012: Convective modes for significant severe thunderstorms in the contiguous United States. Part I: Storm classification and climatology. *Wea. Forecasting*, **27**, 1114–1135, https://doi.org/10.1175/WAF-D-11-00115.1.

Song, F., Z. Feng, L. R. Leung, R. A. Houze Jr., J. Wang, J. Hardin, and C. R. Homeyer, 2019: Contrasting spring and summer large-scale environments associated with mesoscale convective systems over the U.S. Great Plains. *J. Climate*, **32**, 6749–6767, https://doi.org/10.1175/JCLI-D-18-0839.1.

——, L. R. Leung, Z. Feng, X. Chen, and Q. Yang, 2022: Observed and projected changes of large-scale environments conducive to spring MCS initiation over the US Great Plains. *Geophys. Res. Lett.*, **49**, e2022GL098799, https://doi.org/10.1029/2022GL098799.

Trier, S. B., and C. A. Davis, 2002: Influence of balanced motions on heavy precipitation within a long-lived convectively generated vortex. *Mon. Wea. Rev.*, **130**, 877–899, https://doi.org/10.1175/1520-0493(2002)130<0877:IOBMOH>2.0.CO;2.

Varble, A., H. Morrison, and E. Zipser, 2020: Effects of underresolved convective dynamics on the evolution of a squall line. *Mon. Wea. Rev.*, **148**, 289–311, https://doi.org/10.1175/MWR-D-19-0187.1.

Whitaker, J. S., and C. A. Davis, 1994: Cyclogenesis in a saturated environment. *J. Atmos. Sci.*, **51**, 889–908, https://doi.org/10.1175/1520-0469(1994)051%3C0889:CIASE%3E2.0.CO;2.

Woollings, T., B. Hoskins, M. Blackburn, and P. Berrisford, 2008: A new Rossby wave–breaking interpretation of the North Atlantic Oscillation. *J. Atmos. Sci.*, **65**, 609–626, https://doi.org/10.1175/2007JAS2347.1.

You, Z., and Y. Deng, 2023: A hierarchical dissection of multiscale forcing on the springtime mesoscale convective systems in the United States. *J. Climate*, **36**, 39–54, https://doi.org/10.1175/JCLI-D-22-0150.1.

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——, ——, Y. Ming, and W. Dong, 2024: A multiscale assessment of the springtime U.S. mesoscale convective systems in the NOAA GFDL AM4. *Climate Dyn.*, **62**, 4017–4030, https://doi.org/10.1007/s00382-024-07114-4.

Zhao, S., and Y. Deng, 2020: Nonmodal growth of atmospheric disturbances relevant to the East Asian pressure surge in boreal winter. *Climate Dyn.*, **54**, 3077–3089, https://doi.org/10.1007/s00382-020-05157-x.

——, ——, and R. X. Black, 2018: An intraseasonal mode of atmospheric variability relevant to the U.S. hydroclimate in boreal summer: Dynamic origin and East Asia connection. *J. Climate*, **31**, 9855–9868, https://doi.org/10.1175/JCLI-D-18-0206.1.

——, ——, and W. Li, 2020: A nonmodal instability perspective of the declining northern midlatitude synoptic variability in boreal summer. *J. Climate*, **33**, 1177–1192, https://doi.org/10.1175/JCLI-D-18-0799.1.

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